UNIT 6

# CHEMICAL REACTIVITY

## Learning objectives:

- Identify metals, non-metals, and metalloids with their general properties.
- Understand the reactivity of alkali and alkaline earth metals, and their uses.
- Learn the position of metals and non-metals in the periodic table.

## 6.1 Introduction to Metals and Non-Metals

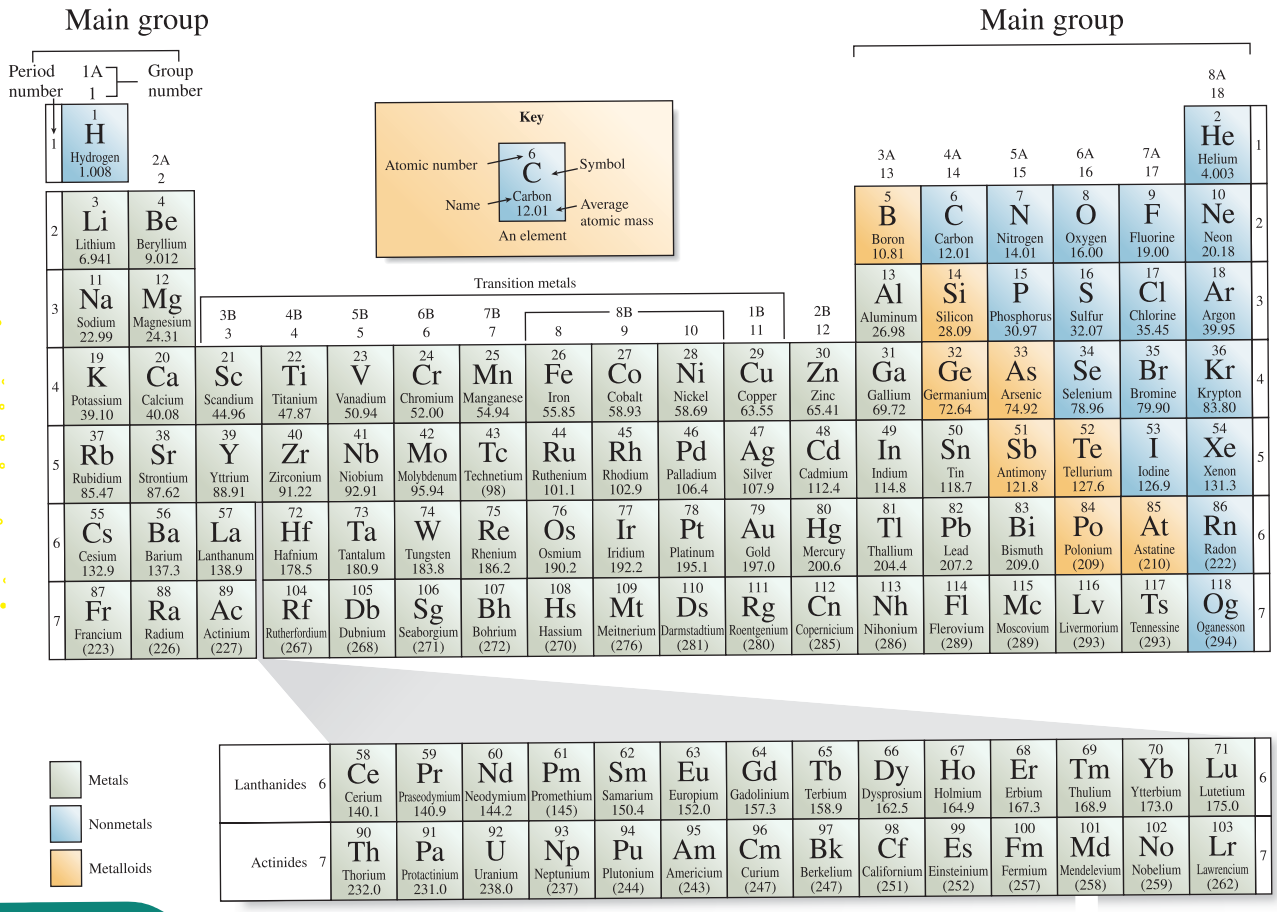
Elements, as you studied in the previous chapter, are classified into metals, non-metals, and metalloids based on their properties.

- **Metals:** They are shiny, and good conductors of heat and electricity. **Examples:** Iron (Fe), Copper (Cu).

- **Non-Metals:** They are dull, and poor conductors of heat and electricity. **Examples:** Oxygen (O), Carbon (C).

- **Metalloids:** These are the elements with properties between metals and non-metals. **Examples:** Silicon (Si), Boron (B).

All the elements are present in the periodic table given below:



## More Properties of Metals and Non-Metals

- **Metals:**

o They have high melting and boiling points.

o They are malleable, and ductile, which means they can be turned into sheets and wires.

- **Non-Metals:**

o They have low melting and boiling points.

o They are brittle, which means they can break easily.

# 6.2 Reactivity of Metals

## Reactivity of Alkali and Alkaline Earth Metals

· **Alkali Metals (Group IA):** Alkali metals are in the Group IA of the modern periodic table. They are very reactive, and vigorously react with water.

**Example:** Sodium (Na) reacts vigorously with water to form sodium hydroxide (NaOH) and hydrogen gas ($H_2$).

$$2Na + 2H_2O \rightarrow 2NaOH + H_2$$

- **Alkaline Earth Metals (Group IIA):** They are also reactive but not as much as alkali metals.

**Example:** Calcium (Ca) reacts with water to form calcium hydroxide ($Ca(OH)_2$) and hydrogen gas ($H_2$).

$$2Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$$

### Common Uses of Reactive Metals

- Sodium is used in streetlights.
- Magnesium is used in fireworks.
- Calcium is used in cement production.

## 6.3 Non-Metals

### Important Properties of Non-Metals

- They are poor conductors of heat and electricity.
- They tend to form acidic oxides when combined with oxygen.
- They gain electrons easily in chemical reactions.

### Some Common Non-Metals and Their Uses

- Halogens (e.g., Chlorine, Fluorine) are used in disinfectants and plastics.
- Oxygen (O) is used in respiration and combustion processes.

Non-metals

# 6.4 Periodic table and reactivity trends

## Position of Metals and Non-Metals in the Periodic Table

- Metals are on the left side and centre.
- Non-metals are on the right side.
- Metalloids are found between metals and non-metals.

Metal | Metalloid | Nonmetal

| | 1<br>1A | 2<br>2A | 3<br>3B | 4<br>4B | 5<br>5B | 6<br>6B | 7<br>7B | 8 | 9<br>8B | 10 | 11<br>1B | 12<br>2B | 13<br>3A | 14<br>4A | 15<br>5A | 16<br>6A | 17<br>7A | 18<br>8A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 H 1.008 | | | | | | | | | | | | | | | | | 2 He 4.002602 |
| 2 | 3 Li 6.94 | 4 Be 9.0121832 | | | | | | | | | | | 5 B 10.81 | 6 C 12.011 | 7 N 14.007 | 8 O 15.999 | 9 F 18.998403164 | 10 Ne 20.1798 |
| 3 | 11 Na 22.98976928 | 12 Mg 24.305 | | | | | | | | | | | 13 Al 26.9815386 | 14 Si 28.085 | 15 P 30.973761999 | 16 S 32.06 | 17 Cl 35.45 | 18 Ar 39.948 |
| 4 | 19 K 39.0983 | 20 Ca 40.078 | 21 Sc 44.955909 | 22 Ti 47.867 | 23 V 50.9415 | 24 Cr 51.9962 | 25 Mn 54.938044 | 26 Fe 55.845 | 27 Co 58.933194 | 28 Ni 58.6934 | 29 Cu 63.546 | 30 Zn 65.38 | 31 Ga 69.723 | 32 Ge 72.631 | 33 As 74.921596 | 34 Se 78.972 | 35 Br 79.904 | 36 Kr 83.798 |
| 5 | 37 Rb 85.4678 | 38 Sr 87.62 | 39 Y 88.90584 | 40 Zr 91.224 | 41 Nb 92.90637 | 42 Mo 95.95 | 43 Tc (98) | 44 Ru 101.07 | 45 Rh 102.90550 | 46 Pd 106.42 | 47 Ag 107.8682 | 48 Cd 112.414 | 49 In 114.818 | 50 Sn 118.711 | 51 Sb 121.760 | 52 Te 127.60 | 53 I 126.90447 | 54 Xe 131.294 |
| 6 | 55 Cs 132.90545197 | 56 Ba 137.328 | 57-71 Lanthanides | 72 Hf 178.49 | 73 Ta 180.94788 | 74 W 183.84 | 75 Re 186.207 | 76 Os 190.23 | 77 Ir 192.217 | 78 Pt 195.085 | 79 Au 196.966570 | 80 Hg 200.592 | 81 Tl 204.38 | 82 Pb 207.2 | 83 Bi 208.98040 | 84 Po (209) | 85 At (210) | 86 Rn (222) |
| 7 | 87 Fr (223) | 88 Ra (226) | 89-103 Actinides | 104 Rf (267) | 105 Db (262) | 106 Sg (271) | 107 Bh (272) | 108 Hs (270) | 109 Mt (276) | 110 Ds (281) | 111 Rg (280) | 112 Cn (285) | 113 Uut (284) | 114 Fl (289) | 115 Uup (288) | 116 Lv (293) | 117 Uus (294) | 118 Uuo (294) |

Transition Metals

Inner Transition Metals

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 57 La 138.90548 | 58 Ce 140.116 | 59 Pr 140.90766 | 60 Nd 144.242 | 61 Pm (145) | 62 Sm 150.36 | 63 Eu 151.964 | 64 Gd 157.25 | 65 Tb 158.92535 | 66 Dy 162.500 | 67 Ho 164.93033 | 68 Er 167.259 | 69 Tm 168.93422 | 70 Yb 173.055 | 71 Lu 174.9668 |
| 89 Ac (227) | 90 Th 232.0377 | 91 Pa 231.03588 | 92 U 238.02891 | 93 Np (237) | 94 Pu (244) | 95 Am (243) | 96 Cm (247) | 97 Bk (247) | 98 Cf (251) | 99 Es (252) | 100 Fm (257) | 101 Md (258) | 102 No (259) | 103 Lr (262) |

## Reactivity Trends in Groups and Periods

- **In Groups:** Groups are the vertical columns of the periodic table. Reactivity increases as you go down the group for alkali metals because the outer electron is farther from the nucleus and easier to lose.

- **In Periods:** Periods are the horizontal rows of periodic table. Reactivity of metals decreases as you move from left to right, while the reactivity of non-metals increases as you move from left to right.

# EXERCISES:

## Fill in the blanks.

**1.** _____ is an example of a non-metal.

**2.** Alkali metals are located in Group _____ of the periodic table.

**3.** The element _____ is commonly used in fireworks.

**4.** _____ is used to make streetlights.

**5.** Non-metals generally have _____ melting and boiling points.

**6.** Metals are generally good conductors of _____ and _____.

**Tick(✓) for True and cross (x) for False statements.**

**1.** Non-metals are good conductors of electricity. ☐

**2.** Calcium is more reactive than sodium. ☐

**3.** The reactivity of alkali metals decreases as you go down the group. ☐

**4.** Oxygen is a non-metal. ☐

**5.** Metalloids are located between metals and non-metals in the periodic table. ☐

**6.** Magnesium is a reactive metal. ☐

**7.** Halogens are found in Group 17 of the periodic table. ☐

**8.** Non-metals are usually shiny and malleable. ☐

## Choose the correct answer.

**1. Which of the following is a property of metals?**

a) Poor conductor of electricity

b) Shiny and malleable

c) Can form acidic oxides

d) Tend to gain electrons

**2. Which group contains the most reactive metals?**

a) Group IA (Alkali metals)

b) Group IIA (Alkaline earth metals)

c) Group VIIA (Halogens)

d) Group VIIIA (Noble gases)

**3. Which of the following is a non-metal?**

a) Sodium b) Copper

c) Oxygen d) Iron

**4. What happens to the reactivity of alkali metals as you go down the group?**

a) It increases b) It decreases

c) It stays the same d) It disappears

**5. What is an example of a metalloid?**

a) Calcium b) Silicon

c) Chlorine d) Nitrogen

**6 . Non-metals are typically:**

a) Ductile b) Malleable

c) Brittle d) Shiny

**7. Which metal is commonly used in fireworks?**

a) Sodium b) Magnesium

c) Calcium d) Iron

**8. Halogens are most commonly used in:**

a) Construction b) Electronics

c) Disinfectants d) Food production

## Answer the Following Questions:

### Short-Answer Questions.

1. **What is element?**
2. **Write the general properties of metals and non-metals.**
3. **Write the differences between alkali metals and alkaline earth metals.**
4. **Write the common uses of reactive metals.**
5. **Where is the position of metals, non-metals and metalloid in the periodic table?**

### Long-Answer Questions.

1. **How does the reactivity of metals and non-metals change across different periods and groups in the periodic table?**

## Create a Periodic Table Model

### ACTIVITY

Students will design a chart of the periodic table, colour-coding metals, non-metals, and metalloids. They will label the groups (e.g., Alkali Metals, Halogens), list properties and examples of each category, and demonstrate how reactivity changes across groups and periods.

### Critical Thinking

Why do alkali metals become more reactive as you move down the group? How does this trend affect their chemical reactions with water?

## Work sheet

**1. Count up the number of atoms on each side of the equation below and write this in the chart.**

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

| Atom | Reactant side | Product side |
|---|---|---|
| Carbon | | |
| Hydrogen | | |
| Oxygen | | |

## Element of Elements

From your knowledge of the fundamental particles of an atom, you have learned that an element has three subatomic particles. Solve for the atomic mass, atomic number, number of protons, neutrons, and electrons for each element given below. The first one is done for you as an example.

### Math and Chemistry Together!

| Element | Symbol | Atomic Number | Atomic Mass | Proton | Neutron | Electron |
|---|---|---|---|---|---|---|
| Lithium | **Li** | 3 | 6.9 | 3 | 4.0 | 3 |
| Nickel | **Ni** | | 58.7 | | | 28 |
| Gold | **Au** | 79 | | | | |
| Uranium - 238 | **U** | | | 92 | 146 | |
| Iodine | **I** | | | | | 53 |
| Barium | **Ba** | 56 | 126.9 | | 81 | |

## Metals, Nonmetals and Metalloids

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| H | | | | | | | | | | | | | | | | | He |
| Li | Be | | | | | | | | | | | B | C | N | O | F | Ne |
| Na | Mg | | | | | | | | | | | Al | Si | P | S | Cl | Ar |
| K | Ca | Sc | Ti | V | Cr | Mn | Fe | Co | Ni | Cu | Zn | Ga | Ge | As | Se | Br | Kr |
| Rb | Sr | Y | Zr | Nb | Mo | Tc | Ru | Rh | Pd | Ag | Cd | In | Sn | Sb | Te | I | Xe |
| Cs | Ba | La | Hf | Ta | W | Re | Os | Ir | Pt | Au | Hg | Tl | Pb | Bi | Po | At | Rn |
| Fr | Ra | Ac | Rf | Db | Sg | Bh | Hs | Mt | Ds | Rg | Cn | Nh | Fl | Mc | Lv | Ts | Og |
| | | | Ce | Pr | Nd | Pm | Sm | Eu | Gd | Tb | Dy | Ho | Er | Tm | Yb | Lu | |
| | | | Th | Pa | U | Np | Pu | Am | Cm | Bk | Cf | Es | Fm | Md | No | Lr | |

Most elements are metals. Elements to the right of the periodic table are non-metals.
Elements: chemical substances that cannot be broken down any smaller.

**2. Identify each element as a metal, nonmetal, or metalloid.**

| | |
|---|---|
| Copper (Cu) | |
| Oxygen (O) | |
| Hydrogen (H) | |
| Potassium (K) | |
| Phosphorous (P) | |
| Helium (He) | |
| Aluminum (Al) | |
| Calcium (Ca) | |
| Magnesium (Mg) | |
| Chlorine (Cl) | |